

DBMS-Unit-1

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Mid-1 Topics

DBMS and it's applications
Integrity
Constraints
ER Models
Banking Database Arxchitecture
DDL
DML
DCL
Commands
Etc....

A database-management system (DBMS) is a collection of interrelated data and a set of programs to access those data. The primary goal of a DBMS is to provide a way to store and retrieve database information that is both convenient and efficient.

Database-System Applications:

1. Enterprise Information
2. Banking and Finance
3. Universities
4. Airlines:
5. Telecommunication

Enterprise Information

- Sales: For customer, product, and purchase information.
- Accounting: For payments, receipts, account balances, assets and other accounting information.
- Human resources: For information about employees, salaries, payroll taxes, and benefits, and for generation of paychecks.
- Manufacturing: For management of the supply chain and for tracking production of items in factories, inventories of items in warehouses and stores, and orders for items.
- Online retailers: For sales data noted above plus online order tracking, generation of recommendation lists, and maintenance of online product evaluations.

Banking and Finance

- Banking: For customer information, accounts, loans, and banking transactions.
- Credit card transactions: For purchases on credit cards and generation of monthly statements.
- Finance: For storing information about holdings, sales, and purchases of financial instruments such as stocks and bonds; also for storing real-time market data to enable online trading by customers and automated trading by the firm.

- Universities: For student information, course registrations, and grades (in addition to standard enterprise information such as human resources and accounting).
- Airlines: For reservations and schedule information. Airlines were among the first to use databases in a geographically distributed manner.
- Telecommunication: For keeping records of calls made, generating monthly bills, maintaining balances on prepaid calling cards, and storing information about the communication networks.

Purpose of Database Systems:

This typical file-processing system is supported by a conventional operating system. The system stores permanent records in various files, and it needs different application programs to extract records from, and add records to, the appropriate files.

Before database management systems (DBMSs) were introduced, organizations usually stored information in such systems. Keeping organizational information in a file-processing system has a number of major disadvantages:

- Data redundancy and inconsistency
- Difficulty in accessing data
- Data isolation
- Integrity problems

- Atomicity problems
- Concurrent-access anomalies.
- Security problems.

These difficulties, among others, prompted the development of database systems. In what follows, we shall see the concepts and algorithms that enable database systems to solve the problems with file-processing systems.

View Of Data

Data Abstraction:

A database system is a collection of interrelated data and a set of programs that allow users to access and modify these data. A major purpose of a database system is to provide users with an abstract view of the data. That is, the system hides certain details of how the data are stored and maintained.

Since many database-system users are not computer trained, developers hide the complexity from users through several levels of abstraction, to simplify users' interactions with the system:

- Physical level.
- Logical level.
- View level

Physical level. The lowest level of abstraction describes how the data are actually stored. The physical level describes complex low-level data structures in detail.

Logical level. The next-higher level of abstraction describes what data are stored in the database, and what relationships exist among those data. The logical level thus describes the entire database in terms of a small number of relatively simple structures.

Database administrators, who must decide what information to keep in the database, use the logical level of abstraction.

View level. The highest level of abstraction describes only part of the entire database.

The view level of abstraction exists to simplify their interaction with the system. The system may provide many views for the same database.

1.3 View of Data

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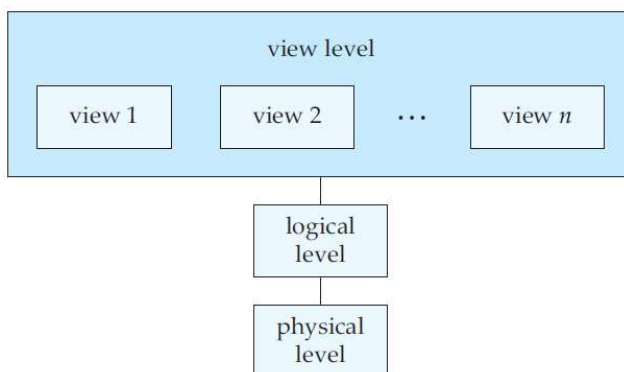


Figure 1.1 The three levels of data abstraction.

```

type instructor = record
    ID : char (5);
    name : char (20);
    dept_name : char (20);
    salary : numeric (8,2);
end;

```

This code defines a new record type called *instructor* with four fields. Each field has a name and a type associated with it. A university organization may have several such record types, including

- *department*, with fields *dept_name*, *building*, and *budget*
- *course*, with fields *course_id*, *title*, *dept_name*, and *credits*
- *student*, with fields *ID*, *name*, *dept_name*, and *tot_cred*

At the physical level, an *instructor*, *department*, or *student* record can be described as a block of consecutive storage locations. The compiler hides this level

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At the logical level, each such record is described by a type definition, as in the previous code segment, and the interrelationship of these record types is defined as well. Programmers using a programming language work at this level of abstraction. Similarly, database administrators usually work at this level of abstraction.

Finally, at the view level, computer users see a set of application programs that hide details of the data types. At the view level, several views of the database are defined, and a database user sees some or all of these views. In addition

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Instances and Schemas:

Databases change over time as information is inserted and deleted.

The collection of information stored in the database at a particular moment is called an instance of the database.

The overall design of the database is called the database schema. Schemas are changed infrequently, if at all.

Database systems have several schemas, partitioned according to the levels of abstraction.

The physical schema describes the database design at the physical level, while the logical schema describes the database design at the logical level.

A database may also have several schemas at the view level, sometimes called subschemas, that describe different views of the database.

Data Models

Underlying the structure of a database is the data model: a collection of conceptual tools for describing data, data relationships, data semantics, and consistency constraints.

A data model provides a way to describe the design of a database at the physical, logical, and view levels.

The data models can be classified into four different categories

- Relational Model.
- Entity-Relationship Model
- Object-Based Data Model
- Semi structured Data Model

Database Languages

A database system provides a **data-definition language** to specify the database schema and a **data-manipulation language** to express database queries and updates

1.4.1 Data-Manipulation Language

A **data-manipulation language (DML)** is a language that enables users to access or manipulate data as organized by the appropriate data model. The types of access are:

- Retrieval of information stored in the database
- Insertion of new information into the database
- Deletion of information from the database
- Modification of information stored in the database

There are basically two types:

- **Procedural DMLs** require a user to specify *what* data are needed and *how* to get those data.
- **Declarative DMLs** (also referred to as **nonprocedural DMLs**) require a user to specify *what* data are needed *without* specifying how to get those data.

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A **query** is a statement requesting the retrieval of information. The portion of a DML that involves information retrieval is called a **query language**. Although technically incorrect, it is common practice to use the terms *query language* and *data-manipulation language* synonymously.

There are a number of database query languages in use, either commercially or experimentally. We study the most widely used query language, SQL, in Chap-

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1.4.2 Data-Definition Language

We specify a database schema by a set of definitions expressed by a special language called a **data-definition language (DDL)**. The DDL is also used to specify additional properties of the data.

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We specify the storage structure and access methods used by the database system by a set of statements in a special type of DDL called a **data storage and definition** language. These statements define the implementation details of the database schemas, which are usually hidden from the users.

The data values stored in the database must satisfy certain **consistency constraints**. For example, suppose the university requires that the account balance of a department must never be negative. The DDL provides facilities to specify such constraints. The database system checks these constraints every time the

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- Domain Constraints.
- Referential Integrity.
- Assertions.
- Authorization ---->
 1. read authorization
 2. insert authorization
 3. update authorization
 4. delete authorization

Relational Databases

A relational database is based on the relational model and uses a collection of tables to represent both data and the relationships among those data. It also includes a DML and DDL.

1.5.1 Tables Each table has multiple columns and each column has a unique name. Figure 1.2 presents a sample relational database comprising two tables: one shows details of university instructors and the other shows details of the various university departments

<i>ID</i>	<i>name</i>	<i>dept_name</i>	<i>salary</i>
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table

<i>dept_name</i>	<i>building</i>	<i>budget</i>
Comp. Sci.	Taylor	100000
Biology	Watson	90000
Elec. Eng.	Taylor	85000
Music	Packard	80000
Finance	Painter	120000
History	Painter	50000
Physics	Watson	70000

(b) The *department* table

Figure 1.2 A sample relational database.

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The relational model is an example of a record-based model. Record-based models are so named because the database is structured in fixed-format records of several types.

1.5.2 Data-Manipulation Language

The SQL query language is nonprocedural. A query takes as input several tables (possibly only one) and always returns a single table. Here is an example of an SQL query that finds the names of all instructors in the History department:

```
select instructor.name
from instructor
where instructor.dept_name = 'History';
```

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Queries may involve information from more than one table. For instance, the following query finds the instructor ID and department name of all instructors associated with a department with budget of greater than \$95,000.

```
select instructor.ID, department.dept_name
from instructor, department
where instructor.dept_name= department.dept_name and
      department.budget > 95000;
```

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1.5.3 Data-Definition Language

SQL provides a rich DDL that allows one to define tables, integrity constraints, assertions, etc.

For instance, the following SQL DDL statement defines the *department* table:

```
create table department
(dept_name char (20),
 building char (15),
 budget numeric (12,2));
```

Execution of the above DDL statement creates the *department* table with three columns: *dept_name*, *building*, and *budget*, each of which has a specific data type associated with it. We discuss data types in more detail in Chapter 3. In addition, the DDL statement creates the data dictionary which contains metadata for the database.

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1.5.4 Database Access from Application Programs

actions must be written in a host language, such as C, C++, or Java, with embedded SQL queries that access the data in the database. Application programs are programs that are used to interact with the database in this fashion

To access the database, DML statements need to be executed from the host language. There are two ways to do this:

- By providing an application program interface (set of procedures) that can be used to send DML and DDL statements to the database and retrieve the results.

The Open Database Connectivity (ODBC) standard for use with the C language is a commonly used application program interface standard. The Java Database Connectivity(JDBC) standard provides corresponding features to the Java language.

- By extending the host language syntax to embed DML calls within the host language program. Usually, a special character prefaces DML calls, and a preprocessor, called the DML precompiler, converts the DML statements to normal procedure calls in the host language.

1.6.3 The Entity-Relationship Model

The entity-relationship (E-R) data model uses a collection of basic objects, called *entities*, and *relationships* among these objects. An entity is a “thing” or “object” in the real world that is distinguishable from other objects. For example, each person is an entity, and bank accounts can be considered as entities.

Entities are described in a database by a set of attributes. For example, the attributes dept name, building, and budget may describe one particular department in a university, and they form attributes of the department entity set

A relationship is an association among several entities. For example, a member relationship associates an instructor with her department. The set of all entities of the same type and the set of all relationships of the same type are termed an entity set and relationship set, respectively.

The overall logical structure (schema) of a database can be expressed graphically by an entity-relationship (E-R) diagram. There are several ways in which to draw these diagrams. One of the most popular is to use the Unified Modeling Language (UML). In the notation we use, which is based on UML, an E-R diagram is represented as follows:

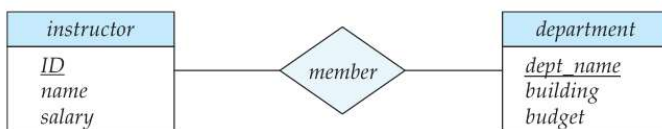


Figure 1.3 A sample E-R diagram.

- Entity sets are represented by a rectangular box with the entity set name in the header and the attributes listed below it.
- Relationship sets are represented by a diamond connecting a pair of related entity sets. The name of the relationship is placed inside the diamond.

Data Storage and Querying

The functional components of a database system can be broadly divided into the storage manager and the query processor components. The storage manager is important because databases typically require a large amount of storage space.

Since the main memory of computers cannot store this much information, the information is stored on disks. Data are moved between disk storage and main memory as needed. Since the movement of data to and from disk is slow relative to the speed of the central processing unit, it is imperative that the database system structure the data so as to minimize the need to move data between disk and main memory.

The query processor is important because it helps the database system to simplify and facilitate access to data. The query processor allows database users to obtain good performance while being able to work at the view level

1.7.1 Storage Manager

The storage manager is the component of a database system that provides the interface between the low-level data stored in the database and the application programs and queries submitted to the system.

The storage manager is responsible for the interaction with the file manager. The raw data are stored on the disk using the file system provided by the operating system. The storage manager translates the various DML statements into low-level file-system commands.

the storage manager is responsible for storing, retrieving, and updating data in the database. The storage manager components include:

- **Authorization and integrity manager**, which tests for the satisfaction of integrity constraints and checks the authority of users to access data.
- **Transaction manager**, which ensures that the database remains in a consistent (correct) state despite system failures, and that concurrent transaction executions proceed without conflicting.
- **File manager**, which manages the allocation of space on disk storage and the data structures used to represent information stored on disk.
- **Buffer manager**, which is responsible for fetching data from disk storage into main memory, and deciding what data to cache in main memory. The buffer manager is a critical part of the database system, since it enables the database to handle data sizes that are much larger than the size of main memory.
- **Data files**, which store the database itself.
- **Data dictionary**, which stores metadata about the structure of the database, in particular the schema of the database.
- **Indices**, which can provide fast access to data items. Like the index in this textbook, a database index provides pointers to those data items that hold a particular value. For example, we could use an index to find the *instructor* record with a particular *ID*, or all *instructor* records with a particular *name*. Hashing is an alternative to indexing that is faster in some but not all cases.

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1.7.2 The Query Processor

The query processor components include:

- **DDL interpreter**, which interprets DDL statements and records the definitions in the data dictionary.
- **DML compiler**, which translates DML statements in a query language into an evaluation plan consisting of low-level instructions that the query evaluation engine understands.

A query can usually be translated into any of a number of alternative evaluation plans that all give the same result. The DML compiler also performs **query optimization**; that is, it picks the lowest cost evaluation plan from among the alternatives.

- **Query evaluation engine**, which executes low-level instructions generated by the DML compiler.

1.8 Transaction Management

one department account (say A) is debited and another department account (say B) is credited. Clearly, it is essential that either both the credit and debit occur, or that neither occur. That is, the funds transfer must happen in its entirety or not at all. This all-or-none requirement is called **atomicity**.

It is essential that the execution of the funds transfer preserve the consistency of the database. That is, the value of the sum of the balances of A and B must be preserved. This correctness requirement is called **consistency**

Finally, after the successful execution of a funds transfer, the new values of the balances of accounts A and B must persist, despite the possibility of system failure. This persistence requirement is called **durability**

A **transaction** is a collection of operations that performs a single logical function in a database application. Each transaction is a unit of both atomicity and consistency.

⁴ Finally, when several transactions update the database concurrently, the consistency of data may no longer be preserved, even though each individual transaction is correct. It is the responsibility of the **concurrency-control manager** to control the interaction among the concurrent transactions, to ensure the consistency of the database. The **transaction manager** consists of the concurrency-control manager and the recovery manager.

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1.9 Database Architecture

Components of a database system and the connections among them

The architecture of a database system is greatly influenced by the underlying computer system on which the database system runs. Database systems can be centralized, or client-server, where one server machine executes work on behalf of multiple client machines. Database systems can also be designed to exploit parallel computer architectures. Distributed databases span multiple geographically separated machines.

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Most users of a database system today are not present at the site of the database system, but connect to it through a network. We can therefore differentiate between **client** machines, on which remote database users work, and **server** machines, on which the database system runs.

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Database applications are usually partitioned into two or three parts, as in Figure 1.6. In a **two-tier architecture**, the application resides at the client machine, where it invokes database system functionality at the server machine through

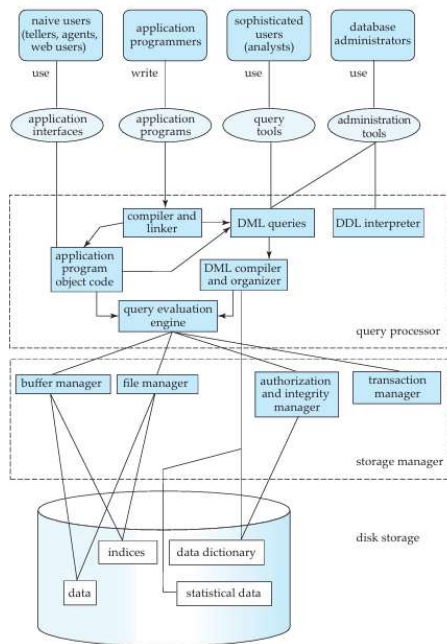


Figure 1.5 System structure.

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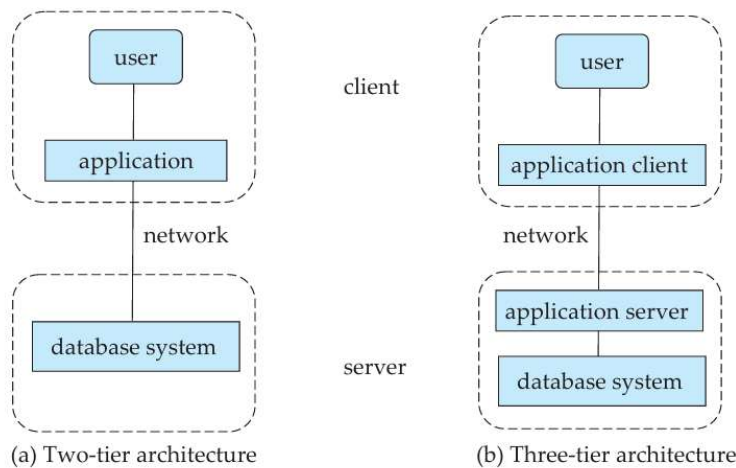


Figure 1.6 Two-tier and three-tier architectures.

Data Mining and Information Retrieval

The term data mining refers loosely to the process of semiautomatically analyzing large databases to find useful patterns. data mining attempts to discover rules and patterns from data. data mining deals with "knowledge discovery in databases." Some types of knowledge discovered from a database can be represented by a set of rules.

Specialty Databases

Several application areas for database systems are limited by the restrictions of the relational data model. As a result, researchers have developed several data models to deal with these application domains, including object-based data models and semi structured data models.

Object-Based Data Models:

This led to the development of an object-oriented data model that can be seen as extending the E-R model with notions of encapsulation, methods (functions), and object identity. Inheritance, object identity, and encapsulation (information hiding), with methods to provide an interface to objects, are among the key concepts of object-oriented programming that have found applications in data modeling. The object-oriented data model also supports a rich type system, including structured and collection types.

The major database vendors presently support the object-relational data model, a data model that combines features of the object-oriented data model and relational data model.

Semi structured Data Models:

Semi structured data models permit the specification of data where individual data items of the same type may have different sets of attributes. This is in contrast with the data models mentioned earlier, where every data item of a particular type must have the same set of attributes.

Database Users and Administrators

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A primary goal of a database system is to retrieve information from and store new information into the database. People who work with a database can be categorized as database users or database administrators.

1.12.1 Database Users and User Interfaces

There are four different types of database-system users, differentiated by the way they expect to interact with the system.

Different types of user interfaces have been designed for the different types of users.

- Naive users
- Application programmers
- Sophisticated users
- Specialized users

• Naive users are unsophisticated users who interact with the system by invoking one of the application programs that have been written previously.

The typical user interface for naive users is a forms interface, where the user can fill in appropriate fields of the form. Naive users may also simply read reports generated from the database

- **Application programmers** are computer professionals who write application programs. Application programmers can choose from many tools to develop user interfaces. **Rapid application development (RAD)** tools are tools that enable an application programmer to construct forms and reports with minimal programming effort.

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- **Sophisticated users** interact with the system without writing programs. Instead, they form their requests either using a database query language or by using tools such as data analysis software. Analysts who submit queries to explore data in the database fall in this category.

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- **Specialized users** are sophisticated users who write specialized database applications that do not fit into the traditional data-processing framework.

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1.12.2 Database Administrator

One of the main reasons for using DBMSs is to have central control of both the data and the programs that access those data. A person who has such central control over the system is called a **database administrator (DBA)**. The functions of a DBA include:

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- Schema definition
- Storage structure and access-method definition.
- Schema and physical-organization modification
- Granting of authorization for data access
- Routine maintenance